Math 105L **Lab 8 Part 1**

Semilog Plots & Modeling Exponential Growth

Exploring Semilog Transformations: $y = 3 \cdot 2^t$

In the spreadsheet demonstration, we begin with the exponential function $y(t) = 3 \cdot 2^t$ and analyze the transformation obtained by taking the natural logarithm of the output, $\ln y$:

t	$y = 3 \cdot 2^t$	$\ln y$	**Spreadsheet Notes / Verification**
0	3	1.0986	$\ln(3) \approx 1.09861$
1	6	1.7918	$\ln(6) \approx 1.79176$ ($\Delta = +0.693$)
2	12	2.4849	$\ln(12) \approx 2.48491$ ($\Delta = +0.693$)
3	24	3.1781	$\ln(24) \approx 3.17805$ ($\Delta = +0.693$)
4	48	3.8712	$\ln(48) \approx 3.87120$ ($\Delta = +0.693$)
5	96	4.5643	$\ln(96) \approx 4.56435$ ($\Delta = +0.693$)

- **Cartesian Graph (y vs. t):** Rapid upward curve displaying classic exponential growth.
- **Semilog Graph ($\ln y$ vs. t):** A **perfect straight line** with constant slope $m \approx 0.693$ and vertical intercept $b \approx 1.0986$.

Board Derivation: Converting Exponential Growth to a Linear Form

Why does taking the natural logarithm of an exponential function linearize the relationship?

$$\begin{aligned} y &= 3 \cdot 2^t \\ \ln y &= \ln\left(3 \cdot 2^t\right) \\ &= \ln 3 + \ln\left(2^t\right) \quad \text{using } \ln(AB) = \ln A + \ln B \\ &= \ln 3 + t \cdot \ln 2 \quad \text{using } \ln(A^p) = p \ln A \\ &= (\ln \mathbf{2}) \cdot \mathbf{t} + \ln \mathbf{3} \end{aligned}$$

Letting $Y = \ln y$, this has the standard slope-intercept form $Y = mt + b$:

$$\text{Slope } m = \ln 2 \approx 0.69315, \qquad \text{Intercept } b = \ln 3 \approx 1.09861$$

Back-Transformation: Reconstructing the Exponential from a Trendline

The Reverse Procedure (Recovering Model Parameters):

Suppose a linear regression on semilog axes gives the equation $\ln y = 0.693t + 1.1$. How do we recover the original exponential model $y(t) = P_0 a^t$ (or $P_0 e^{rt}$)?

$$\begin{aligned} \ln y &= 0.693t + 1.1 \\ e^{\ln y} &= e^{0.693t+1.1} \\ y &= e^{1.1} \cdot e^{0.693t} \quad \text{using } e^{A+B} = e^A \cdot e^B \\ y &\approx 3.004 \cdot \left(e^{0.693}\right)^t \\ y &\approx \mathbf{3.004} \cdot (\mathbf{1.9997})^{\mathbf{t}} \approx \mathbf{3} \cdot \mathbf{2}^{\mathbf{t}} \end{aligned}$$

Real-World Case Study: U.S. Historical Population (1790–1900)

U.S. Census Data & Semilog Regression

We model U.S. historical population growth from the initial census in 1790 ($t = 0$) through 1900 ($t = 110$ years):

Year	**Years Since 1790 (t)**	**Population (millions)**	ln(**pop**)	**Exponential Model: $P(t)$**
1790	0	3.9	1.3610	4.19
1800	10	5.3	1.6677	5.52
1810	20	7.2	1.9741	7.26
1820	30	9.6	2.2618	9.56
1830	40	12.9	2.5572	12.59
1840	50	17.1	2.8391	16.57
1850	60	23.2	3.1442	21.82
1860	70	31.4	3.4468	28.72
1870	80	39.8	3.6839	37.81
1880	90	50.2	3.9160	49.78
1890	100	62.9	4.1415	65.54
1900	110	76.0	4.3307	86.29

Extracting the Continuous Exponential Model

- **Spreadsheet Trendline on** ln(**pop**) **vs.** t**:**

$$\ln(\text{pop}) = 0.0275 \cdot t + 1.43$$

- **Back-Transformation via Exponentiation:**

$$\text{pop}(t) = e^{0.0275t+1.43} = e^{1.43} \cdot e^{0.0275t} \approx \mathbf{4.19 \cdot e^{0.0275t}}$$

- **Spreadsheet Formula Implemented:** In cell `E2`: `=4.19*exp(0.0275*B2)`
- **Interpretation of Model Parameters:**
  - **Base Population ($t = 0$, 1790):** $P_0 = e^{1.43} \approx 4.19$ million people (very close to actual 3.9M).
  - **Continuous Growth Rate:** $r = 0.0275 = 2.75\%$ per year throughout the 19th century.
  - **Doubling Time:**

$$t_{\text{double}} = \frac{\ln 2}{r} = \frac{0.69315}{0.0275} \approx \mathbf{25.2} \text{ years}$$

Summary: The Semilog Workflow for Exponential Data

1. Transform raw (t, y) pairs into $(t, \ln y)$.
2. Plot $\ln y$ vs. t. A linear pattern confirms that exponential modeling is valid.
3. Fit a linear trendline: $\ln y = mt + b$.
4. Recover the natural parameters: initial value $P_0 = e^b$, continuous growth rate $r = m$.